



Section 4: Cell Organelles

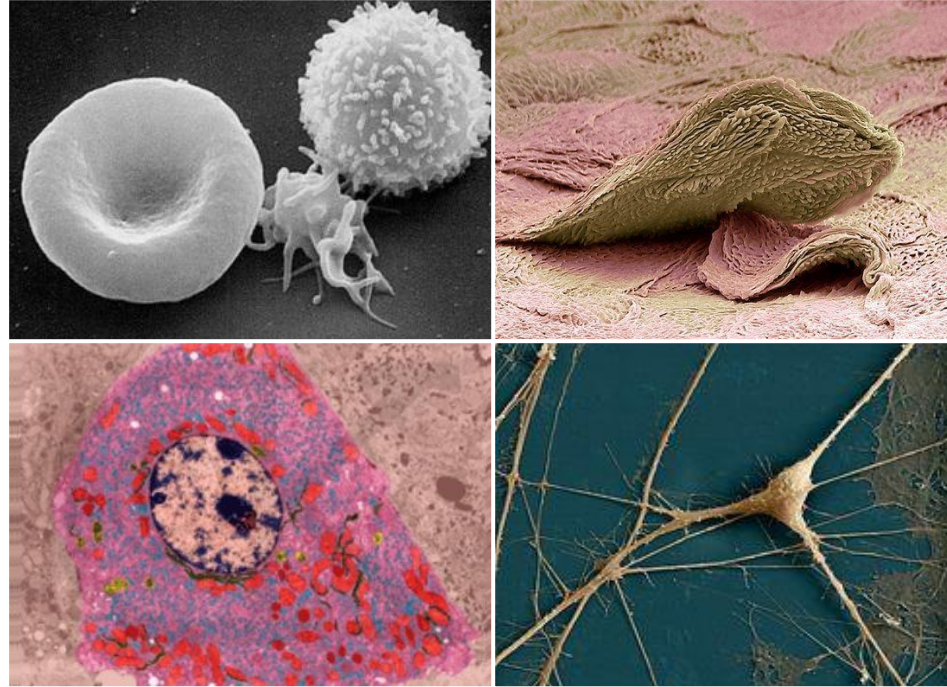
Learning Targets

- I can explain how organelles interact as a system to help cells maintain homeostasis.
- I can match each organelle to its correct physical description and function.

Cell Organelles: Background

- **Organelles**: small parts of a cell that have a specific function
- When organelles all work together as a system, the cell maintains homeostasis.

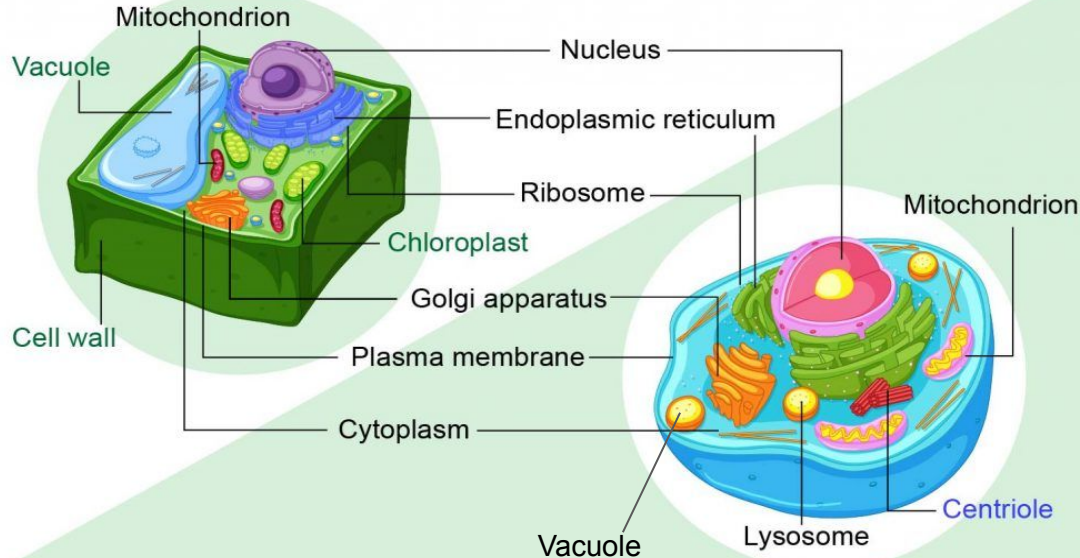
Review: What is homeostasis?



Organelles in Plant & Animal Cells

Warm Up Day 1

PLANT CELL



ANIMAL CELL

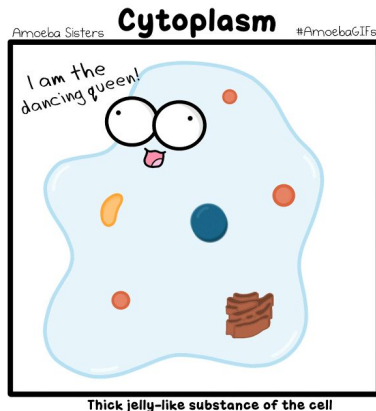
Each organelle in plant and animal cells has a specific function.

Can you name organelles in each cell based on the diagram?

Cytoplasm & Cytoskeleton

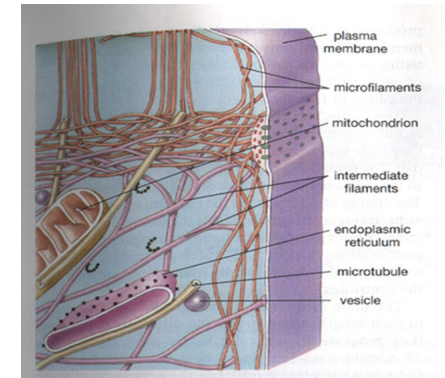
Cytoplasm

- Physical Description: A jellylike substance
- Function: contains dissolved nutrients and surrounds all of the organelles



Cytoskeleton

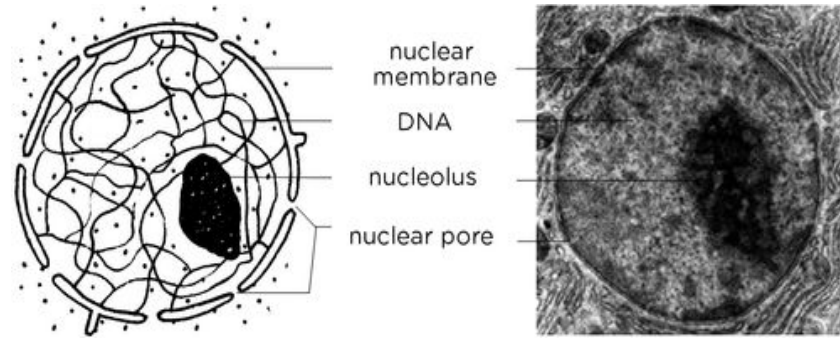
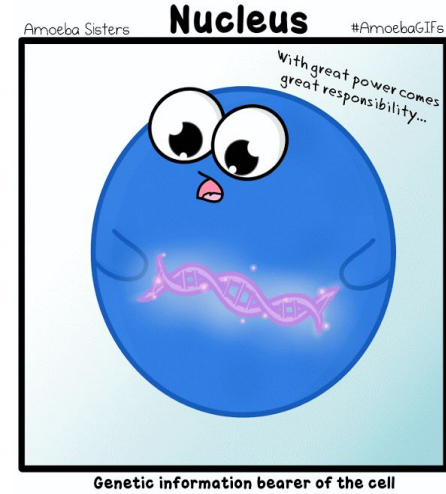
- Physical Description: A network of proteins
- Function: provides structure for the cell and anchors organelles in certain spots



Nucleus

Nucleus

- Physical Description:
 - Enclosed inside nuclear envelope, which has pores allowing molecules to pass in and out
- Function:
 - Storehouse for genetic material/DNA
 - Two major demands-
 1. DNA is protected
 2. DNA must be available for use to make proteins



Mitochondria

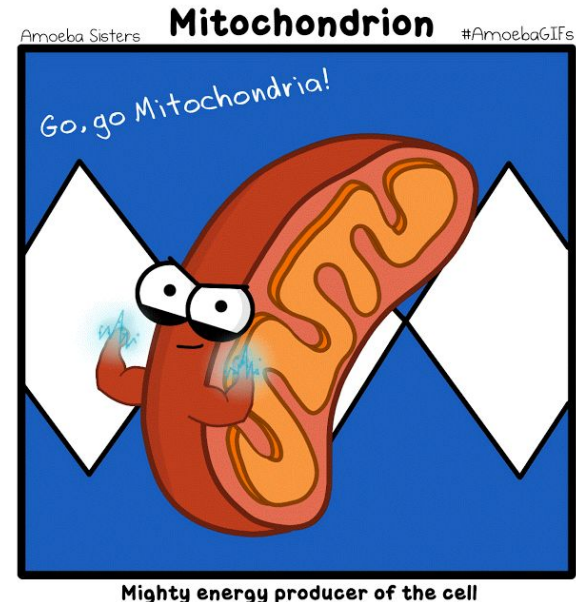
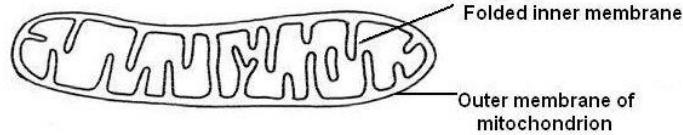
Mitochondria

- Physical Description:

- Bean shaped with a double membrane
- Have their own ribosomes and DNA
 - suggests mitochondria were free-living prokaryotes taken in by larger cells

- Function:

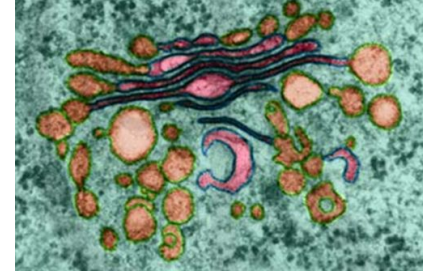
- Supply energy to the cell, using cellular respiration



Golgi Apparatus

Golgi Apparatus

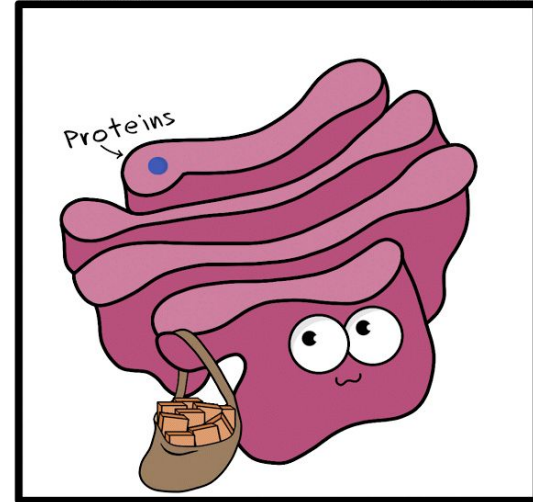
- Physical Description:
 - Closely layered stacks of membrane enclosed sacs
- Function:
 - Process, sort, and deliver proteins in vesicles
 - Vesicles: small membrane-bound sacs that separate some material from the rest of the cytoplasm



Golgi Apparatus

Amoeba Sisters

#AmoebaGIFs

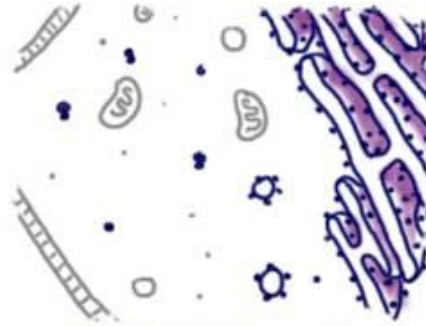


Post office of the cell

Ribosomes

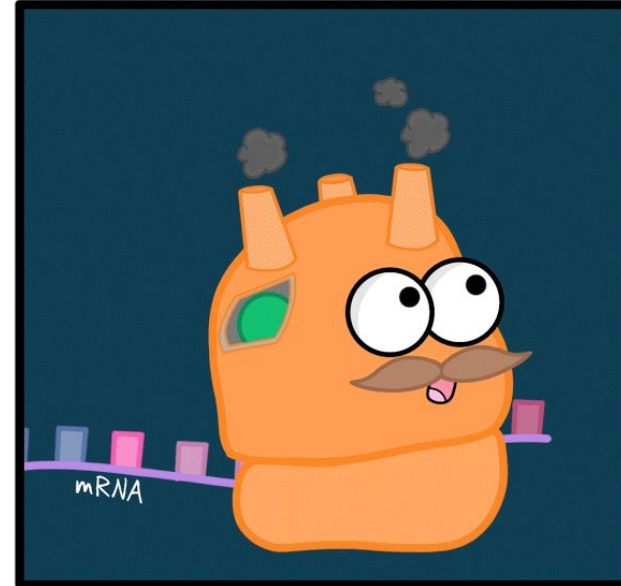
Ribosomes

- Physical Description:
 - Made up of protein and RNA
- Function:
 - Uses messenger RNA and transfer RNA to synthesize (make) proteins



RIBOSOMES CAN BE
FOUND FLOATING FREE
AND ATTACHED TO THE ER.

Amoeba Sisters **Ribosomes** #AmoebaGIFs

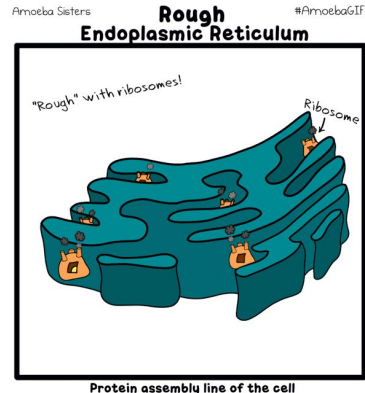
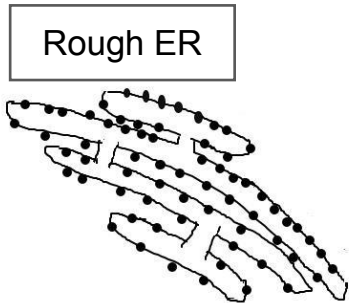


Protein synthesizers of the cell

Rough ER & Smooth ER

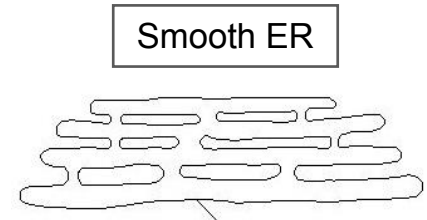
Rough ER

- Physical Description: interconnected network of folded membranes studded with ribosomes
- Function: production of **proteins** occurs on the surface and inside

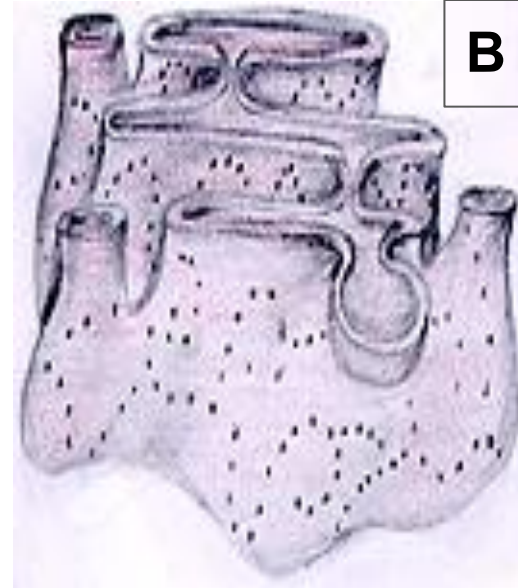
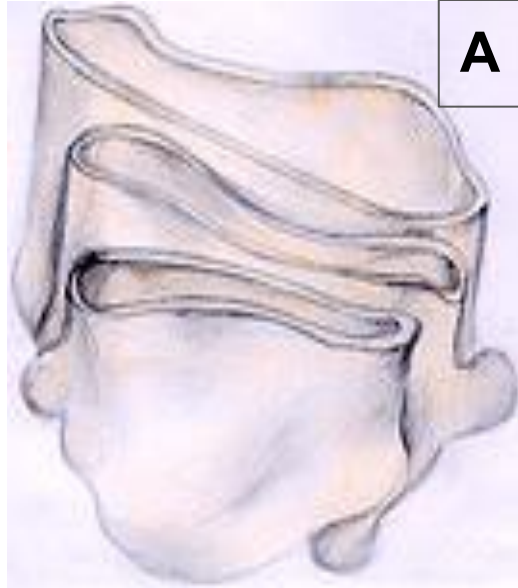


Smooth ER

- Physical Description: interconnected network of folded membranes **WITHOUT** ribosomes
- Function: production of **lipids** occurs on the inside and breaks down drugs and alcohol



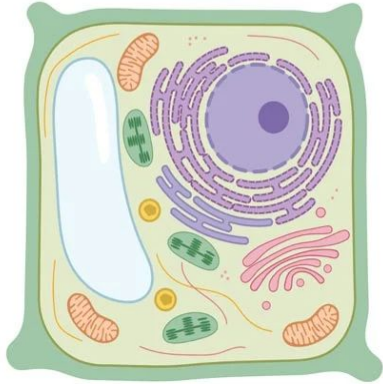
Rough ER & Smooth ER: Closer Day 1



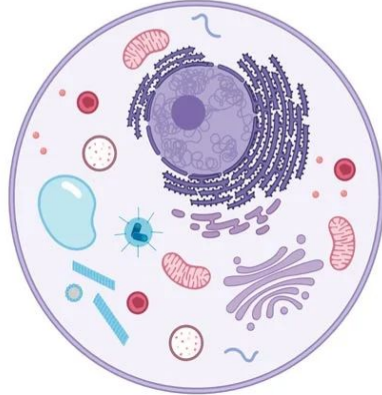
1. Which image shows Rough ER? Which image shows Smooth ER?
2. Why do they look different?

Section 4: Warm Up Day 2

Plant Cell



Animal Cell



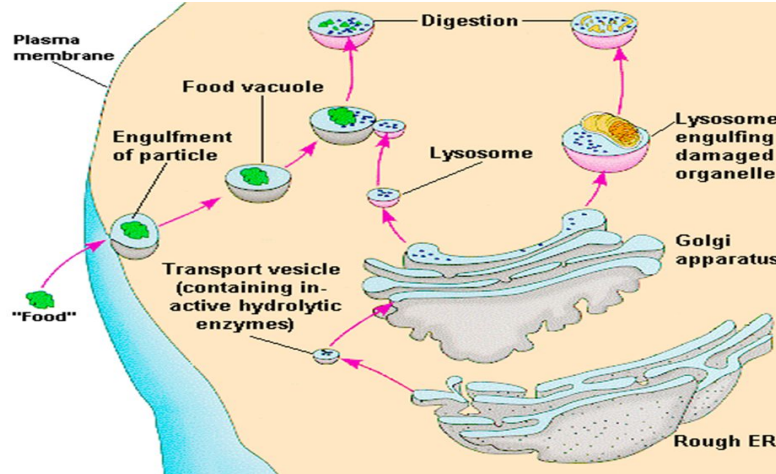
Organelles interact as a system within a cell to maintain homeostasis. Each one plays an important role.

1. Support this statement with 3 examples of organelles. Name each organelle and explain why it is important to the cell.
1. If one organelle fails to perform its role, what would happen to the cell?

Lysosome

Lysosome

- Physical Description:
 - Membrane bound organelles that contain enzymes
- Function:
 - Defends cell from bacteria and viruses
 - Breaks down damaged or worn out cell parts



Lysosome

#AmoebaGIFs

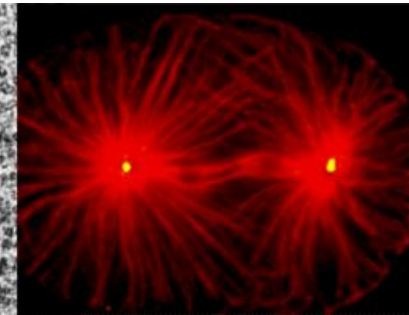
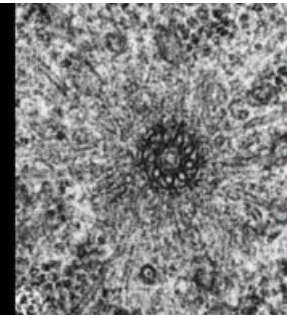
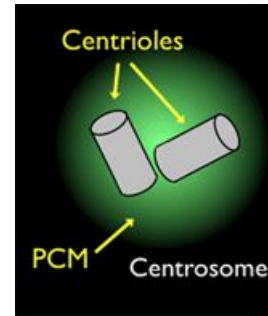
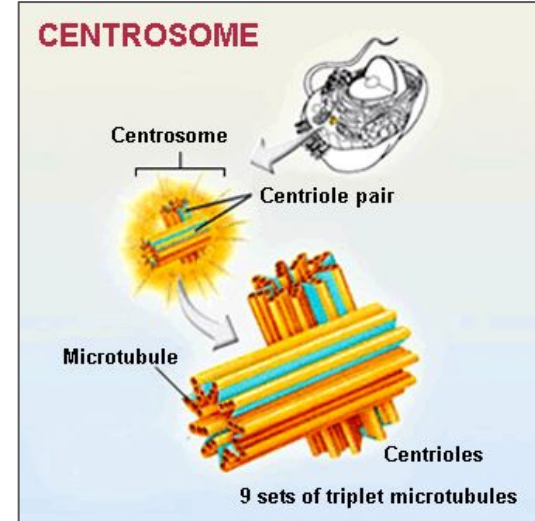


Enzyme-packed wrecking balls of the cell

Centrosome

Centrosome

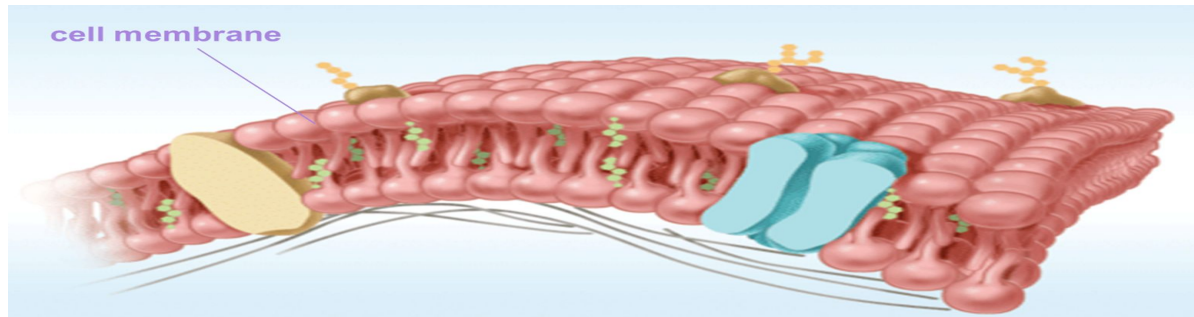
- Physical Description:
 - **Found in animal cells** (not plant cells)
 - Small region that produces microtubules arranged in a circle called centrioles
- Function:
 - Helps the cell divide and organizes microtubules into cilia and flagella



Cell Membrane

Cell Membrane

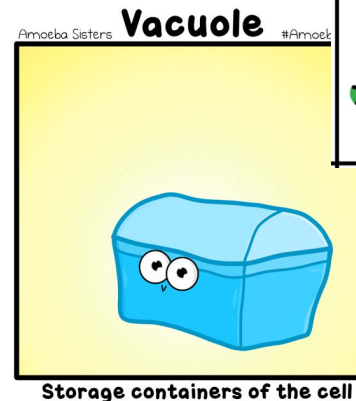
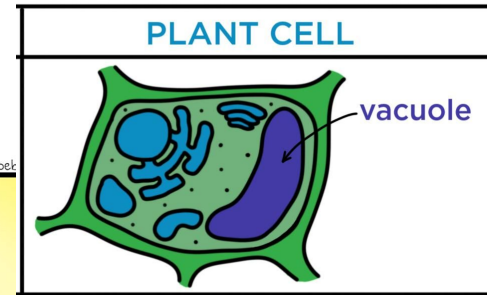
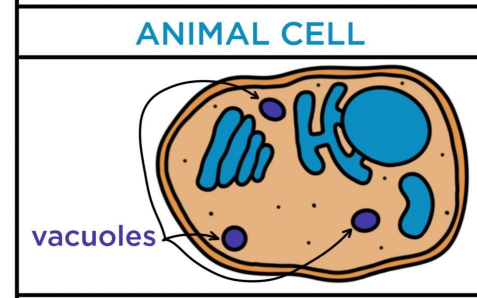
- Physical Description:
 - Composed of special lipids called phospholipids
 - Surrounds the cell
- Function:
 - Protects the cell and allows things (nutrients, food, molecules) in and out of the cell



Vacuoles

Vacuoles

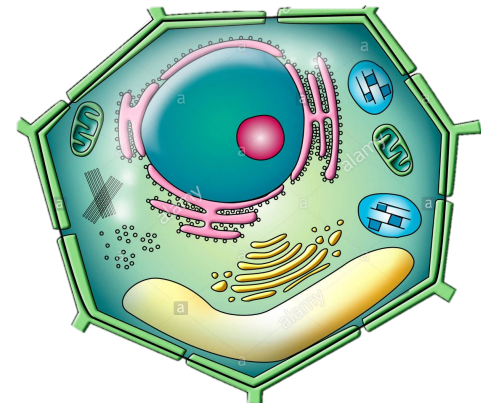
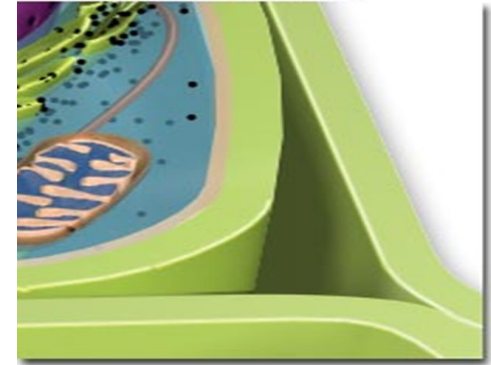
- Physical Description:
 - Fluid filled sac found in both plant and animal cells
 - **Animal cells** - many small vacuoles
 - **Plant cells** - a single large central vacuole
- Function:
 - Used for storage
 - Plant only: filled with watery fluid that strengthens and supports the cell



Cell Wall

Cell Wall

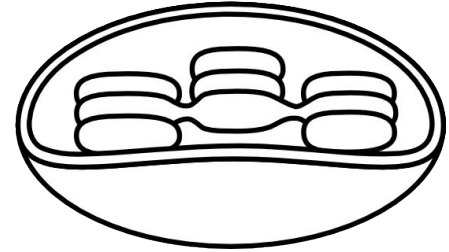
- Physical Description:
 - Rigid layer that surrounds the cell membrane
 - **Found in plants, algae, fungi and most bacteria** (not in animal cells)
 - Plants and algae: made of cellulose
 - Fungi: made of chitin
- Function:
 - gives protection, support, and shape to the cell



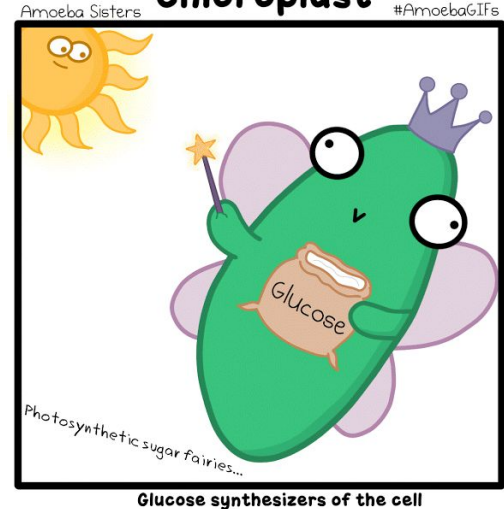
Chloroplast

Chloroplast

- Physical Description:
 - **Only found in plants and green algae** (not in animal cells)
 - Have their own ribosomes and DNA
 - Led scientists to believe that (like mitochondria) they were originally free-living prokaryotes
- Function:
 - carry out photosynthesis



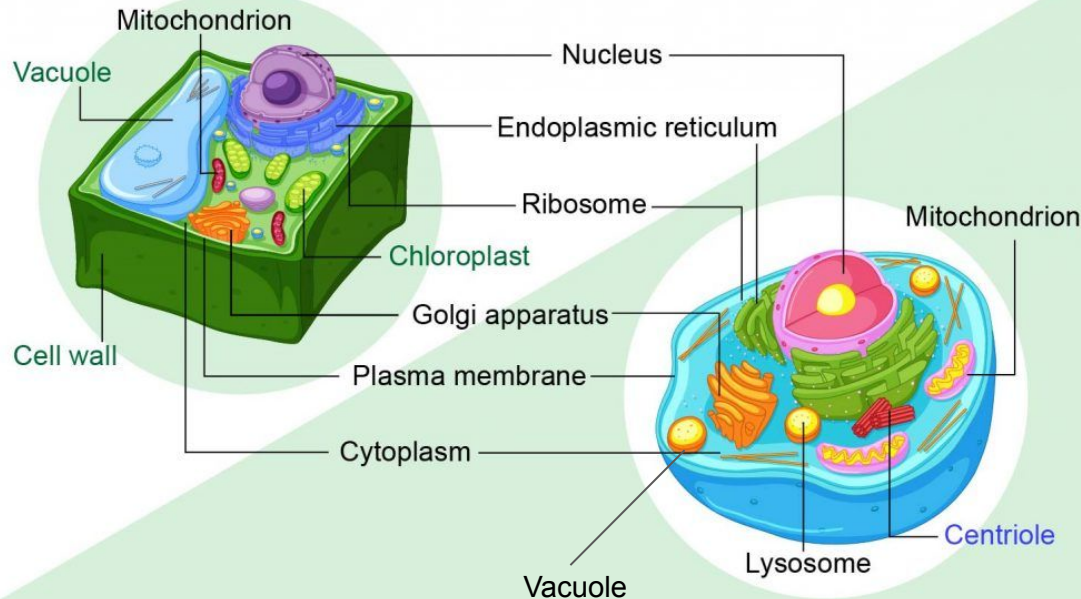
Chloroplast



Section 4: Closer Day 2

Use the diagram below to answer the questions.

PLANT CELL



ANIMAL CELL

1. Name 2 organelles that are only in the plant cell.
2. Name 1 organelle that is only in the animal cell.
3. Describe how vacuoles are different between plant and animal cells.
4. Name at least 5 organelles that are found in BOTH plant and animal cells.